Lecture 17: Finite Square Barriers, Tunneling, and Finite Wells

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Lecture 16 introduced probability current and scattering from a single potential step. We now add a second interface. The two-interface problem produces two characteristically quantum effects: transmission resonances above a finite barrier and tunneling below it. We then compare scattering and bound states of a finite square well.

Questions.

- How do the stationary solutions change when a potential has two interfaces?
- Why can a finite barrier become perfectly transparent at special energies?
- How can a particle cross a barrier even when its energy is below the barrier height?
- Which parameters control the exponential suppression of tunneling?
- How do scattering states and bound states differ in a finite square well?

Lecture roadmap. We first review current and the potential step. We then solve the finite square barrier for energies above and below its height, emphasizing boundary matching, interference, resonance, and tunneling. Finally, we use the corresponding finite-well problem to contrast continuous scattering states with discrete bound states.

Review from Lecture 16

For a stationary state, the probability current is

$$j = \frac{\hbar}{2mi}\left(\phi^* \frac{\mathrm{d}\phi}{\mathrm{d}x} - \phi \frac{\mathrm{d}\phi^*}{\mathrm{d}x}\right). \tag{1}$$

The right- and left-moving plane waves carry currents

$$j[Ae^{ikx}] = \frac{\hbar k}{m}|A|^2, \qquad j[Be^{-ikx}] = -\frac{\hbar k}{m}|B|^2. \tag{2}$$

For incidence from the left, reflection and transmission are defined by current ratios:

$$\boxed{R = \frac{|j_{\mathrm{refl}}|}{j_{\mathrm{in}}}, \qquad T = \frac{j_{\mathrm{trans}}}{j_{\mathrm{in}}}, \qquad R + T = 1.} \tag{3}$$

At a semi-infinite step with $0 < E < V_0$, the wave function penetrates into the higher-potential region as a decaying exponential. That exponential carries no current, and there is no second boundary that could convert it back into a propagating wave. Consequently,

$$\boxed{R = 1, \qquad T = 0 \qquad \text{for a semi-infinite step with } 0 < E < V_0.} \tag{4}$$

For $E > V_0$, waves propagate on both sides. Their wave numbers are different, so a reflected component is generally needed to satisfy the matching conditions.

Questions.

Question 1. Why can a decaying tail occur beyond a semi-infinite step with $0 < E < V_0$ even though the transmission probability is zero?

Answer and explanation. The wave function need not vanish in the classically forbidden region. However, a single real decaying exponential has zero probability current. Because the step extends to $+\infty$, there is no second interface at which the remaining amplitude can become an outgoing traveling wave. Thus there is penetration but no transmitted flux.

1 The finite square barrier

Consider

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad V_0 > 0. \tag{5}$$

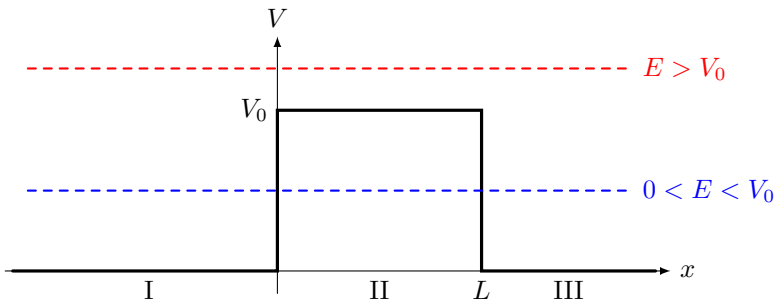


Figure 1: A finite square barrier has interfaces at $x = 0$ and $x = L$.

The potential is zero in Regions I and III, so their wave numbers are the same. The essential difference from a step is the second interface:

potential step:	one interface,
finite barrier:	two interfaces.

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Waves reflected from the two interfaces can interfere. We first consider energies above the barrier and then energies below it.

2 Above-barrier scattering: $E > V_0$

2.1 Stationary solutions in the three regions

Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad q = \frac{\sqrt{2m(E - V_0)}}{\hbar}. \tag{7}$$

The Schrödinger equation becomes

region	equation	solution for left incidence
$x < 0$	$\phi_I'' + k^2\phi_I = 0$	$Ae^{ikx} + Be^{-ikx}$
$0 < x < L$	$\phi_{II}'' + q^2\phi_{II} = 0$	$Ce^{iqx} + De^{-iqx}$
$x > L$	$\phi_{III}'' + k^2\phi_{III} = 0$	Fe^{ikx}

(8)

Thus

$$\boxed{\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Ce^{iqx} + De^{-iqx}, & 0 < x < L, \\ Fe^{ikx}, & x > L. \end{cases}} \tag{9}$$

Region I contains the incident and reflected waves. Both directions must be retained inside the barrier because each interface reflects. Region III has only a right-moving wave because no wave is incident from $+\infty$.

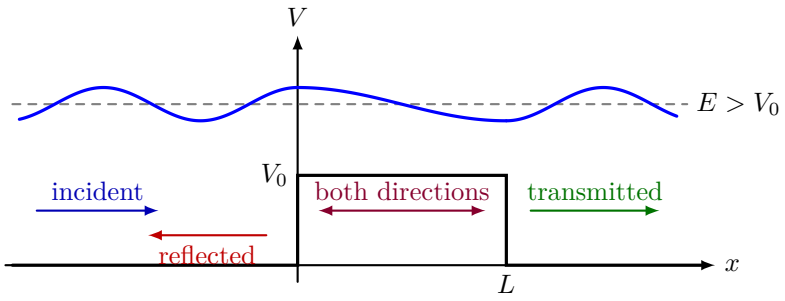


Figure 2: For $E > V_0$, the stationary wave is oscillatory in all three regions.

2.2 Matching at both interfaces

Because the barrier is finite and nonsingular, both ϕ and ϕ' are continuous at $x = 0$ and $x = L$. Equation (9) therefore gives

$$\boxed{\begin{aligned} x = 0: \quad & A + B = C + D, \\ & k(A - B) = q(C - D), \\ x = L: \quad & Ce^{iqL} + De^{-iqL} = Fe^{ikL}, \\ & q(Ce^{iqL} - De^{-iqL}) = kFe^{ikL}. \end{aligned}} \tag{10}$$

These four equations determine the ratios B/A, C/A, D/A, and F/A. Since Regions I and III have the same wave number,

$$R = \left|\frac{B}{A}\right|^2, \qquad T = \left|\frac{F}{A}\right|^2. \tag{11}$$

It is useful to introduce the dimensionless quantities

$$\varepsilon = \frac{E}{V_0}, \qquad g_0^2 = \frac{2mV_0L^2}{\hbar^2}. \tag{12}$$

Solving the matching equations yields

$$\boxed{T = \left[1 + \frac{1}{4\varepsilon(\varepsilon-1)} \sin^2\left(g_0\sqrt{\varepsilon-1}\right)\right]^{-1}, \qquad \varepsilon > 1.} \tag{13}$$

Equivalently,

$$T = \left[1 + \frac{V_0^2}{4E(E-V_0)} \sin^2(qL)\right]^{-1}. \tag{14}$$

2.3 Transmission resonances

The sine term vanishes whenever

$$\boxed{qL = n\pi, \qquad n = 1, 2, 3, \ldots.} \tag{15}$$

At these special energies,

$$\boxed{T = 1, \qquad R = 0.} \tag{16}$$

This is a *transmission resonance*. An integer number of half-wavelengths fits across the barrier because $qL = n\pi$ is equivalent to $L = n\lambda_{II}/2$. The reflected amplitudes associated with multiple paths through the two-interface region cancel in the net reflected wave.

Perfect transmission at resonance is an interference effect. It is not the same as the high-energy limit. When $E \gg V_0$, $q \approx k$, so each interface individually produces little reflection and $T \to 1$. At resonance, by contrast, individual reflected contributions need not be small; their sum vanishes because of their relative phases.

Questions.

Question 2. A finite square barrier has $E > V_0$ and $qL = n\pi$. What are R and T, and what physical mechanism produces this result?

Answer and explanation. The condition makes $\sin^2(qL) = 0$ in the exact transmission formula, so $T = 1$ and $R = 0$. Reflected amplitudes arising from the two interfaces and from repeated internal reflections have phases that cancel the total reflected wave. The potential is still present; its transparency is caused by interference.

3 Below-barrier scattering and tunneling: $0 < E < V_0$

3.1 Evanescent solutions inside a finite barrier

Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad \kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}. \tag{17}$$

In Region II the Schrödinger equation becomes

$$\phi_{II}'' - \kappa^2 \phi_{II} = 0. \tag{18}$$

For incidence from the left, the complete solution is

$$\boxed{\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Ce^{\kappa x} + De^{-\kappa x}, & 0 < x < L, \\ Fe^{ikx}, & x > L. \end{cases}} \tag{19}$$

Neither exponential is discarded inside the finite barrier. The interval $0 < x < L$ does not extend to infinity, so neither term violates normalizability there. Both are generally required to satisfy the matching conditions at the two interfaces.

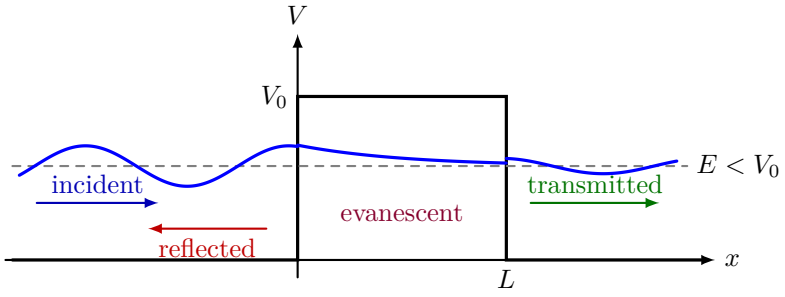


Figure 3: Below a finite barrier, the evanescent wave reaches the second interface and generates an outgoing transmitted wave.

The matching equations have the same form as before after replacing q with $i\kappa$. Using

$$\sin(i\kappa L) = i\sinh(\kappa L), \tag{20}$$

the transmission probability becomes

$$\boxed{T = \left[1 + \frac{1}{4\varepsilon(1-\varepsilon)}\sinh^2\left(g_0\sqrt{1-\varepsilon}\right)\right]^{-1}, \qquad 0 < \varepsilon < 1.} \tag{21}$$

Equivalently,

$$T = \left[1 + \frac{V_0^2}{4E(V_0 - E)}\sinh^2(\kappa L)\right]^{-1}. \tag{22}$$

Although $E < V_0$, this expression is nonzero for finite L. A propagating wave and nonzero probability current appear beyond the classically forbidden region. This is *quantum tunneling*.

The contrast with the semi-infinite step is now clear:

	second interface	transmitted current for $E < V_0$
semi-infinite step	absent	$T = 0$
finite barrier	present	$T > 0$

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3.2 Thick-barrier approximation

When $\kappa L \gg 1$,

$$\sinh(\kappa L) \approx \frac{1}{2}e^{\kappa L}. \tag{24}$$

Equation (21) then gives

$$\boxed{T \approx 16\frac{E}{V_0}\left(1-\frac{E}{V_0}\right)e^{-2\kappa L} \propto e^{-2\kappa L}.} \tag{25}$$

Define the penetration length

$$\ell = \frac{1}{\kappa} = \frac{\hbar}{\sqrt{2m(V_0-E)}}. \tag{26}$$

Then the dominant behavior is

$$\boxed{T \propto e^{-2L/\ell}.} \tag{27}$$

Consequently,

$$\boxed{\begin{array}{ccc} L\uparrow & \Longrightarrow & T\downarrow \text{ exponentially,} \\ V_0-E\uparrow & \Longrightarrow & \kappa\uparrow \Longrightarrow T\downarrow, \\ m\uparrow & \Longrightarrow & \kappa\uparrow \Longrightarrow T\downarrow. \end{array}} \tag{28}$$

Tunneling is therefore extremely sensitive to the barrier width, the energy deficit $V_0 - E$, and the particle mass. Since $\sinh u$ has no nonzero real zeros, a single square barrier has no perfect-transmission resonance below the barrier.

Questions.

Question 3. In the thick-barrier regime, which changes increase the tunneling probability: decreasing L, decreasing m, or moving E closer to V_0 from below?

Answer and explanation. All three changes increase tunneling. They reduce the exponent $2\kappa L$, where $\kappa = \sqrt{2m(V_0-E)}/\hbar$. Because T depends exponentially on this combination, even a modest reduction can produce a large relative increase in transmission.

3.3 Stationary states and wave packets

The formulas above use stationary scattering eigenfunctions, which extend over all space. A localized incident particle is instead represented by a wave packet containing a range of energies. After interacting with the barrier, it separates into reflected and transmitted outgoing packets:

$$\text{incident packet} \longrightarrow \text{reflected packet} + \text{transmitted packet}. \tag{29}$$

For a narrow energy distribution, the probabilities are approximately $R(E_0)$ and $T(E_0)$ at the central energy E_0. Near a sharp resonance, however, different energy components can have noticeably different transmission probabilities, so a finite packet need not transmit perfectly even when $T(E_0) = 1$.

4 Comparing the two energy regions of a barrier

	$E > V_0$	$0 < E < V_0$
Region II solution	oscillatory	evanescent
interior scale	$q = \sqrt{2m(E - V_0)}/\hbar$	$\kappa = \sqrt{2m(V_0 - E)}/\hbar$
transmission	oscillatory in energy	exponentially suppressed
perfect transmission	possible at $qL = n\pi$	absent for one barrier
main phenomenon	resonance	tunneling

The two exact transmission formulas are analytic continuations of one another: when $q = i\kappa$, $\sin(qL)$ becomes $i\sinh(\kappa L)$. The change from oscillatory to evanescent behavior is therefore encoded directly in the Schrödinger equation.

Questions.

Question 4. Why can $T(E)$ oscillate and reach one for $E > V_0$, while it decreases approximately exponentially for $0 < E < V_0$?

Answer and explanation. Above the barrier, the interior wave number q is real. Propagation across the barrier accumulates an oscillatory phase qL, allowing reflected amplitudes to interfere destructively at special energies. Below the barrier, $q = i\kappa$ and the interior solution is evanescent; the amplitude is attenuated across the barrier by a factor controlled by $e^{-\kappa L}$ rather than by a propagating phase.

5 The finite square well

Now reverse the sign of the middle potential:

$$V(x) = \begin{cases} 0, & x < 0, \\ -V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad V_0 > 0. \tag{30}$$

The same two-interface matching logic applies, but the physical behavior depends on whether the energy lies above or below the outside potential, whose zero has been chosen as $V = 0$.

$E > 0$	scattering states	continuous energies
$-V_0 < E < 0$	bound states	discrete energies

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5.1 Positive-energy scattering states

For $E > 0$, define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(E + V_0)}}{\hbar}. \tag{32}$$

The scattering solution for incidence from the left is

$$\phi_E(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Ce^{ik_2 x} + De^{-ik_2 x}, & 0 < x < L, \\ Fe^{ikx}, & x > L. \end{cases} \tag{33}$$

Because $k_2 > k$, the wavelength is shorter inside the attractive well. The wave is oscillatory outside and extends to infinity, so these states belong to a continuous scattering spectrum.

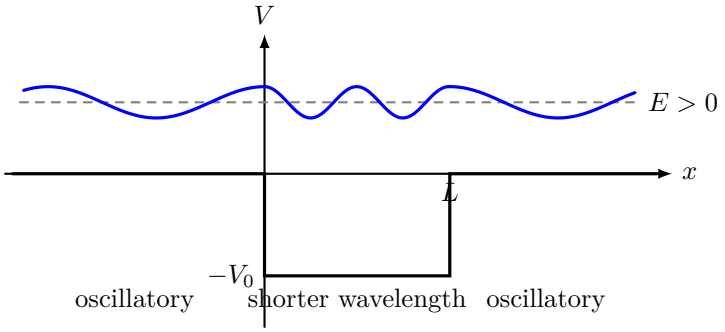


Figure 4: A positive-energy scattering state of a finite square well.

With $\varepsilon = E/V_0$ and $g_0^2 = 2mV_0L^2/\hbar^2$, the transmission probability is

$$\boxed{T(E) = \left[1 + \frac{1}{4\varepsilon(\varepsilon+1)} \sin^2\left(g_0\sqrt{\varepsilon+1}\right)\right]^{-1}.} \tag{34}$$

The well also has transmission resonances:

$$\boxed{k_2 L = n\pi \quad \Longrightarrow \quad T = 1.} \tag{35}$$

At resonance, an integer number of half-wavelengths fits across the well and the net reflected amplitude vanishes.

5.2 Negative-energy bound states

For $-V_0 < E < 0$, define

$$q_b = \frac{\sqrt{2m(E+V_0)}}{\hbar}, \qquad \kappa = \frac{\sqrt{-2mE}}{\hbar}. \tag{36}$$

The wave is oscillatory inside the well but exponential outside. Normalizability selects the decaying exponential on each side:

$$\boxed{\phi_b(x) = \begin{cases} Ae^{\kappa x}, & x < 0, \\ B\cos(q_b x) + C\sin(q_b x), & 0 < x < L, \\ De^{-\kappa(x-L)}, & x > L. \end{cases}} \tag{37}$$

As $x \to -\infty$, $e^{\kappa x} \to 0$; as $x \to +\infty$, $e^{-\kappa(x-L)} \to 0$. The growing exponentials on the two outer half-lines are excluded.

Continuity of ϕ and ϕ' at both boundaries gives four homogeneous linear equations for A, B, C, D. A nonzero solution exists only for special energies. Hence

$$\boxed{\text{normalizability + matching at } x = 0, L \Longrightarrow \text{discrete bound-state energies.}} \tag{38}$$

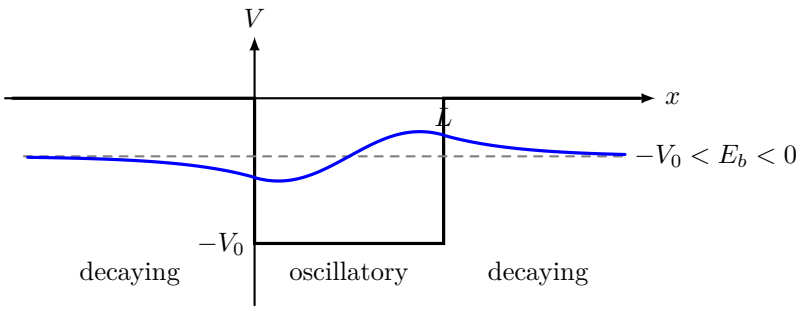


Figure 5: A schematic bound-state eigenfunction of a finite square well.

A negative energy is not unphysical. Energy is measured relative to the chosen zero of potential. Here $E < 0$ means that the state lies below the outside potential and therefore cannot propagate to spatial infinity.

Questions.

Question 5. For a finite square well, how can one determine from the energy and the behavior at infinity whether a state is a scattering state or a bound state?

Answer and explanation. If $E > 0$, the energy exceeds the outside potential, so the outer solutions are oscillatory and the state belongs to the continuous scattering spectrum. If $-V_0 < E < 0$, the outer solutions are exponential. Normalizability keeps only the decaying exponentials, localizing the particle; matching then permits only discrete bound-state energies.

Lecture 17 summary

- A finite barrier has two interfaces, so internal reflections can interfere. For $E > V_0$,

$$T = \left[1 + \frac{V_0^2}{4E(E - V_0)} \sin^2(qL)\right]^{-1}, \qquad q = \frac{\sqrt{2m(E - V_0)}}{\hbar}.$$

- Above-barrier transmission resonances occur when

$$qL = n\pi, \qquad T = 1, \qquad R = 0.$$

- For $0 < E < V_0$, the interior wave is evanescent but a finite barrier still allows tunneling:

$$T = \left[1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(\kappa L)\right]^{-1} > 0.$$

- For a thick barrier,

$$T \propto e^{-2\kappa L}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

  Thus wider barriers, heavier particles, and larger $V_0 - E$ suppress tunneling.
- A finite square well has continuous scattering states for $E > 0$ and discrete bound states for $-V_0 < E < 0$. Bound states oscillate inside the well and decay exponentially outside.

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., the sections on the finite square barrier, quantum tunneling, transmission resonances, and the finite square well. Focus on setting up the regional solutions and applying the matching conditions before using the closed-form transmission formulas.